

Release 1 Product Definition and Scope

1. Product definition

Release 1 is a human-reviewed Document AI operations platform for small independent insurance agencies. It accepts a customer's personal auto insurance declaration page, extracts the most useful policy information, identifies missing or questionable values, gives an insurance employee a clear review workspace, and updates a lightweight customer record only after approval. The product preserves the original document, AI output, reviewer corrections, processing history, and usage information so every important result can be explained and traced.

One-sentence definition: A human-reviewed AI workspace that converts personal auto insurance declaration pages into reliable, traceable customer and policy records.

This should not be positioned as a complete agency management system, insurance CRM, quoting system, or autonomous insurance agent. It is a focused intake and review product that could later connect with those systems.

2. Primary user

Recommended primary user

The primary user should be a **new-business intake specialist or personal-lines account manager at a small independent U.S. insurance agency with approximately 3-20 employees.**

The agency owner may be the buyer, but the intake specialist or account manager is the person who would use the product daily.

This is narrower and more believable than targeting every employee at every type of insurance company.

Current manual workflow

The employee usually:

1. Receives a declaration page from a prospect or existing customer through email or another upload method.
2. Downloads and opens the document.
3. Searches through one or more pages for policy information.
4. Manually enters customer, vehicle, driver, coverage, date, and premium information into an agency system, CRM, spreadsheet, or quoting workflow.
5. Checks whether important information is missing.
6. Contacts the customer when the document is unclear or incomplete.
7. Rechecks values because mistakes can affect a quote or customer record.
8. Keeps limited evidence of who entered or corrected each value.

Main pain

The work is repetitive but still requires attention. Employees repeatedly switch between a PDF and a data-entry screen, interpret different carrier layouts, and verify whether values belong to the policy, a vehicle, or an individual driver.

The problem is not simply that typing is slow. The larger problems are:

- incorrect policy numbers, dates, limits, or deductibles
- missed vehicles or drivers
- inconsistent handling between employees
- slow response to new prospects
- repeated customer follow-up
- limited visibility into what the AI or employee changed
- no safe way to automate without trusting AI too much

Narrow first use case

Release 1 should focus on this situation:

An agency employee receives a current personal auto declaration page from a prospect and needs to turn it into a reviewed customer and policy record before continuing the quoting or onboarding process.

Release 1 should not attempt to quote, recommend coverage, compare prices, bind a policy, or make insurance decisions.

3. Core business problem

Insurance declaration pages contain useful structured information, but they are delivered as documents with layouts that vary by carrier and sometimes by state, product, or document version.

Agency employees must manually locate and re-enter this information. This creates four related costs.

Labor cost

A trained employee spends time reading documents and performing data entry instead of serving customers, preparing quotes, or completing higher-value work.

Error cost

A value that looks reasonable can still be wrong. An incorrect expiration date, deductible, vehicle, or policy limit can cause quoting mistakes, repeated work, customer confusion, or bad internal records.

Delay cost

A prospect may wait while an employee processes documents. Slower intake can delay quoting and reduce the chance that the agency responds before a competitor.

Control and trust cost

Basic extraction tools may return values without showing where they came from. Fully automated AI creates another risk: it can produce a believable but unsupported value.

The product solves this by automating the first pass while keeping a human responsible for the final decision. It should reduce repetitive reading and typing without hiding uncertainty or removing accountability.

4. Golden path

The golden path is the one workflow that must feel complete, understandable, and reliable before Release 1 is considered finished.

1. The staff member signs in.

The user enters a controlled single-agency workspace.

2. The staff member creates or finds a customer.

The user searches for an existing customer or creates a minimal customer record using basic contact information.

3. The staff member uploads a declaration page.

The user selects an English-language U.S. personal auto declaration page in PDF format and associates it with the customer.

4. The system checks the upload.

The system verifies that the file can be opened, is within the supported size and page limits, appears to be the supported document type, and is not an obvious duplicate.

5. The document enters background processing.

The user can leave the page without breaking processing. The document receives a clear status such as Uploaded, Processing, Needs Review, Approved, Rejected, or Failed.

6. The system obtains usable text.

It first attempts direct PDF text extraction. If the document does not contain enough usable text, it uses OCR.

7. The AI produces structured candidate values.

The selected model identifies the supported policy, customer, vehicle, driver, and coverage information.

8. The system validates and explains the result.

It checks formats, missing values, date relationships, repeated values, conflicting values, and whether supporting text can be located in the document.

9. The document enters the review queue.

The reviewer sees which documents are ready, which failed, and which contain important warnings.

10. The reviewer compares the document with the extracted values.

The reviewer can view the source document, inspect supporting evidence, edit incorrect values, mark values as not present, and understand why a field was flagged.

11. The reviewer resolves the document.

The reviewer approves the document, rejects it with a reason, or sends it through a controlled reprocessing attempt.

12. Approved values update the customer record.

Only approved information becomes the active policy information associated with the customer. Existing approved data is not silently overwritten.

13. The system records the complete history.

The audit history shows the upload, processing attempts, AI configuration version, warnings, reviewer edits, approval or rejection, and customer record update.

14. The reviewer sends a simple confirmation when appropriate.

After approval, the reviewer may send a narrow acknowledgement email stating that the document was received and reviewed. The email should not provide insurance advice or claim that coverage has been verified by the carrier.

15. The workflow remains reusable.

A demonstration user can process another controlled sample without developer assistance, database access, or manual cleanup.

5. Recommended release strategy

Strategy A: Document-first release

This strategy delivers:

- customer records
- declaration page upload
- text extraction and OCR
- structured AI extraction
- validation
- uncertainty handling
- human review
- approval and rejection
- audit history
- usage controls

Voice intake is added after the document workflow is stable.

Advantages

- One AI reliability problem is solved at a time.
- Document results can be tested repeatedly using the same files.
- Failed and successful cases can be reproduced.

- The human review workflow receives enough design attention.
- The product reaches a polished state sooner.
- Development and demonstration costs remain easier to control.
- The release directly supports Document AI, OCR, workflow automation, FastAPI, Next.js, human-in-the-loop AI, and insurance automation portfolio services.
- It creates the customer and audit foundations that Voice AI will later use.
- It is easier to explain in interviews because the data flow and failure handling are visible.

Disadvantages

- Release 1 will not yet demonstrate Voice AI.
- The complete future customer journey is not visible immediately.
- The initial marketing story is narrower.

Strategy B: Thin connected workflow

This strategy adds a limited browser voice intake before document upload.

A customer would speak with an AI receptionist, provide basic information, and then continue to the declaration page upload.

Advantages

- The demo may look broader and more impressive at first.
- It immediately shows that Voice AI and Document AI can share a customer record.
- It supports a wider range of Upwork proposal examples.

Disadvantages

- It introduces audio capture, transcription, turn-taking, interruptions, microphone permissions, browser compatibility, voice output, latency, and conversation recovery.
- It adds another AI system with different failure conditions.
- Voice demonstrations are less repeatable than document demonstrations.
- Browser and microphone problems can damage a live Loom or client demonstration.
- Additional provider usage can increase costs.
- The team may polish the voice experience while leaving the more valuable document review workflow incomplete.
- The connection between voice intake and document review may remain shallow if both are built too quickly.
- A weak voice experience can make the whole product feel less production-ready.

Recommendation

Choose Strategy A: Document-first Release 1.

This is the better choice for development complexity, reliability, budget, learning value, and the time required to produce a credible public demo.

The key objective is not to show the largest number of AI features. It is to show that one AI workflow has been designed responsibly from upload through approval.

Release 1 should establish the customer identity, workflow status, audit, AI processing, human review, and cost-control foundations. A small browser voice intake can then become Release 2 and connect to real product records instead of being a disconnected demonstration.

Voice AI remains part of the product vision, but including it in Release 1 would be scope creep.

6. Release 1 scope

Only the following capabilities belong in Release 1.

6.1 Controlled staff access and one agency workspace

User problem it solves

Customer documents contain personal information. A public visitor should not be able to view, modify, or approve records.

Why it belongs in Release 1

Authentication is necessary for a believable operations platform, audit history, reviewer identity, and safe public demonstrations.

Release 1 does not need a complete organization-management system.

Successful version

- Only authenticated users can access customer documents and records.
 - Every important action can be connected to a user.
 - All users belong to one predefined agency workspace.
 - The public demo uses synthetic data and a controlled demo account or controlled access flow.
 - There is no complex invitation, department, or custom permission system.
-

6.2 Minimal customer records

User problem it solves

A document must belong to a real business object. Extracted values are not useful if they remain isolated from the customer they describe.

Why it belongs in Release 1

Customer records create the connection between document processing, future Voice AI, and later CRM features. This is what makes the product one connected SaaS rather than an extraction playground.

Successful version

A user can:

- create a customer
- search for a customer
- view basic customer contact information
- view documents belonging to the customer
- view the latest approved auto policy information
- view important customer-related audit activity

The customer record should remain intentionally small. It is not a full CRM.

6.3 Document intake and lifecycle

User problem it solves

Users need to know whether a document was uploaded successfully, whether it is being processed, and what they should do next.

Why it belongs in Release 1

Processing can take longer than a normal page action. Without a clear lifecycle, users may upload the same document repeatedly or assume that the system is broken.

Successful version

- A PDF can be uploaded and assigned to a customer.
 - Unsupported file types and excessive file sizes are rejected before expensive processing.
 - Every document has one clear current status.
 - The status is visible in customer records, document lists, and the review workspace.
 - Processing continues without requiring the browser page to remain open.
 - Loading, processing, success, failure, rejected, and empty states are designed.
 - The user never has to guess whether the document is waiting, running, finished, or blocked.
-

6.4 Direct text extraction with OCR fallback

User problem it solves

Some PDFs contain selectable text. Others are scans that contain only page images. Applying the same expensive process to every document wastes money and can reduce quality.

Why it belongs in Release 1

A useful insurance document product must support both common PDF conditions. Direct text extraction should be attempted first because it is normally faster and less expensive. OCR should be used only when needed.

Successful version

- The system identifies whether the PDF contains enough usable text.
 - Text-based PDFs avoid unnecessary OCR calls.
 - Readable scanned PDFs can proceed through OCR.
 - Low-quality, incomplete, or unreadable OCR results are clearly flagged.
 - The user can see whether direct extraction or OCR was used.
 - The extracted text is treated as processing evidence, not automatically as approved customer data.
 - Rotated or poorly oriented pages are either handled or reported clearly rather than silently producing bad results.
-

6.5 Structured AI extraction

User problem it solves

Raw PDF or OCR text is still difficult to use. The agency employee needs organized policy information rather than a large block of text.

Why it belongs in Release 1

This is the central AI capability and the main source of portfolio value.

Successful version

- One selected model provider extracts only the supported fields.
- Output that does not match the expected structure is rejected or repaired through a controlled process.
- Missing fields remain missing rather than being guessed.
- Multiple vehicles and drivers remain separate.
- Policy-level information is not incorrectly attached to a specific vehicle.
- The model and extraction instruction version are recorded with the processing attempt.
- The same curated document and the same extraction configuration produce acceptably repeatable results.
- Extraction output remains a candidate result until a human approves it.

Release 1 should support one model provider in production. The internal design should make future replacement possible, but it should not build and maintain several provider integrations.

6.6 Validation, source evidence, and uncertainty signals

User problem it solves

A clean-looking AI result may still be incorrect. Reviewers need help deciding where to spend their attention.

Why it belongs in Release 1

Without this layer, the product is only a formatted model response. Validation and evidence are what make the workflow believable as an operational product.

Successful version

For each important value, the system can communicate:

- whether it was found
- whether supporting text was located
- which document page supports it
- whether its format is valid
- whether it conflicts with another value
- whether the field is normally important but missing
- whether OCR quality may have affected it
- whether a reviewer changed it

The system should use practical labels such as:

- Ready to review
- Missing
- Invalid
- Conflicting
- Low evidence
- OCR concern
- Reviewer corrected

A model-generated confidence percentage must not be presented as proof of correctness.

6.7 Review queue

User problem it solves

Agency employees need to know which documents require attention rather than opening customers one by one.

Why it belongs in Release 1

The queue turns extraction into an operations workflow. It makes the product feel like a tool used for daily work rather than a one-document demo.

Successful version

The reviewer can see:

- documents waiting for review
- documents with critical warnings
- failed documents
- approved and rejected documents
- customer name
- upload time

- processing status
- basic warning summary

Release 1 does not need complex assignment rules, service-level timers, team workload balancing, or approval chains.

6.8 Human review workspace

User problem it solves

Reviewers must be able to confirm values without constantly switching between separate applications or trusting AI blindly.

Why it belongs in Release 1

Human review is a central product capability, not a fallback screen. It is the main safety control for incorrect AI extraction.

Successful version

The workspace provides:

- a readable source document view
- extracted values grouped by policy, customer, vehicles, drivers, and coverages
- visible warnings next to affected fields
- source text and page references where available
- clear separation between the original AI value and a reviewer correction
- the ability to edit a value
- the ability to mark a field as not present
- the ability to acknowledge a warning
- the ability to approve the complete document
- the ability to reject the document with a reason
- the ability to request a controlled reprocessing attempt

The approval action should be unavailable while unresolved critical validation problems remain, unless the reviewer explicitly acknowledges why approval is still valid.

Release 1 does not need field-by-field multi-user approvals or manager sign-off.

6.9 Approval, rejection, and customer record updates

User problem it solves

The system needs a clear boundary between untrusted AI output and business-approved information.

Why it belongs in Release 1

Without an approval boundary, users cannot explain which data is official, when it changed, or whether AI results have already affected the customer record.

Successful version

- AI extraction does not directly update active customer policy data.
 - Approval creates a clear approved policy version.
 - Reviewer corrections are used in the approved version.
 - Existing approved values are not silently overwritten.
 - The user can see which document produced the current policy information.
 - Rejected documents do not update the active customer policy.
 - Reprocessing does not remove an already approved version.
 - A new approved document can replace the active version only through an explicit action.
-

6.10 Audit history and processing versions

User problem it solves

When a value changes, users need to know who changed it, why it changed, and which AI process produced the original result.

Why it belongs in Release 1

Traceability is one of the strongest differences between a serious AI product and a simple upload-and-prompt demo.

It is also valuable for interviews because it demonstrates that AI behavior, reviewer behavior, and business data were treated separately.

Successful version

The audit history records important events such as:

- customer creation or change
- document upload
- duplicate warning
- processing start and completion
- direct text extraction or OCR usage
- model and extraction version used
- processing failure
- retry or reprocessing attempt
- reviewer field correction
- warning acknowledgement
- approval
- rejection and reason
- customer policy update
- confirmation email attempt and outcome

The history should be understandable to a normal user. It should not display only technical logs.

6.11 Safe retries and duplicate protection

User problem it solves

Users may refresh pages, click actions twice, upload the same file again, or retry after a provider timeout. These actions should not corrupt records or create several approved policies.

Why it belongs in Release 1

Duplicate processing is a common production failure. Preventing it demonstrates reliable backend design and creates strong interview material.

Successful version

- Repeating an upload action does not silently create several active policy records.
- An obvious duplicate produces a warning before another paid processing attempt.
- A retry creates a traceable processing attempt.
- Failed attempts remain visible.
- A retry does not overwrite reviewer edits or approved data.
- Repeated approval requests do not apply the same update several times.
- The user receives a clear result when an action has already completed.

6.12 Narrow customer confirmation

User problem it solves

A customer may not know whether the agency received and reviewed the document.

Why it belongs in Release 1

A narrow confirmation closes the workflow and demonstrates a useful business outcome without building a complete messaging platform.

Successful version

- A reviewer can send a predefined acknowledgement after approval.
- The message states that the document was received and processed.
- It does not claim that the policy is active, valid, verified by the carrier, or suitable for the customer.
- The send result is recorded.
- Demo mode prevents arbitrary visitors from sending messages to uncontrolled recipients.
- Failure to send an email does not reverse document approval.

There should be no email campaign builder, template editor, inbox, conversation thread, SMS, or notification preference center.

6.13 Usage and cost controls

User problem it solves

AI, OCR, storage, and public demo usage can produce unexpected costs.

Why it belongs in Release 1

The product is being built under a limited budget, and cost-aware AI architecture is itself valuable portfolio evidence.

Successful version

The product records enough information to show:

- which provider was used
- which model or OCR service was used
- whether OCR was required
- document page count
- processing attempts
- model usage information when supplied by the provider
- a clearly labelled cost estimate when a reliable configured rate is available

The product also supports:

- upload size and page limits
- retry limits
- development and demo usage caps
- restricted live-processing access
- cached or replayed results for selected samples
- a visible warning when processing is disabled because a limit has been reached

Release 1 needs a practical operations view, not an advanced financial reporting product.

6.14 Reusable public demonstration mode

User problem it solves

A public portfolio demo can be damaged by deleted data, provider outages, uncontrolled costs, or visitors entering sensitive information.

Why it belongs in Release 1

The product's immediate purpose includes Loom videos, Upwork proposals, technical interviews, and live demonstrations.

Successful version

- The demo uses synthetic customers and synthetic insurance documents.

- Sample workflows can be reset without database access.
 - At least one sample can run using stored processing results.
 - Live AI processing is limited and clearly marked.
 - Demo visitors cannot view real customer information.
 - Demo data can be restored to a known state.
 - The product explains that it is a demonstration and not an insurance decision system.
 - The core workflow remains demonstrable when an external AI or OCR provider is temporarily unavailable.
-

7. Explicitly excluded features

The following features should not be built in Release 1.

Browser Voice AI

Voice intake belongs in Release 2 after customer identity, document processing, audit history, and status management are stable.

Building it now would introduce audio and conversational failure modes before the core workflow is polished.

Real phone calls

Phone numbers, telephony providers, call recording, regional rules, voicemail, interruptions, call transfer, and call costs create a separate operational system.

Human call transfer

Transfer requires agent availability, routing rules, failure handling, and telephony integration. It has no value until real phone calling exists.

Appointment scheduling integrations

Scheduling is not required to prove document intake and review. It also creates calendar provider and timezone complexity.

Multiple workspaces

Release 1 should operate as one agency workspace. Multi-tenant isolation should be considered in architecture, but workspace creation, switching, onboarding, branding, and management should wait.

Complex team permissions

Release 1 needs authenticated users and actor identity, not custom roles, permission matrices, departments, or approval chains.

Advanced dashboards

A small operational summary is useful. A large analytics dashboard is not. Release 1 should not include trend charts, employee productivity reports, conversion reporting, or executive reporting.

Detailed cost reports

Basic usage visibility and approximate cost controls are required. Cost allocation by team, customer, provider, period, or billing category is not.

Multiple insurance document types

Supporting identification cards, claim forms, loss runs, home declarations, commercial policies, or applications would multiply document rules and test requirements.

Legal document extraction

This would weaken the insurance-specific product story and introduce unrelated fields and expectations.

Document-grounded chat

Chat may look attractive, but it does not improve the golden path. It creates citation, access-control, and conversational accuracy concerns.

Excel export

Export is useful later, but it does not solve the first workflow better than an approved customer record.

Public APIs

An API product requires external authentication, rate limits, versioning, documentation, and support expectations. No external customer needs it in Release 1.

Webhooks

There are no external workflows that require real-time delivery in the first release.

Advanced analytics

Field accuracy trends, reviewer productivity, carrier comparisons, and correction patterns require enough real usage data to be meaningful.

Custom extraction schemas

Custom fields would turn the focused product into a general document extraction platform and make validation much harder.

Multiple model providers

Release 1 should not build provider switching controls or maintain several production integrations. A clean internal boundary and stored normalized results are enough.

Multiple voice providers

Voice is not in Release 1, so voice provider abstraction is unnecessary.

White-labeling

Custom logos, domains, colors, emails, and agency-specific experiences do not improve the first operational workflow.

Complex billing and subscription plans

The immediate product is a portfolio and demonstration system. There is no validated pricing model or customer billing requirement.

Full CRM functionality

Release 1 should not include sales pipelines, activities, tasks, notes, opportunities, campaigns, lead scoring, or general contact management.

Agency management system replacement

The product should not attempt policy servicing, renewals, claims, commissions, quoting, binding, certificates, carrier downloads, or accounting.

Direct carrier verification

Extracting information from a declaration page does not prove that the policy is active or that the carrier confirms the values. Release 1 must not imply otherwise.

8. Supported document and fields

Exact first document type

Release 1 should support:

An English-language U.S. personal auto insurance declaration page supplied as a PDF, containing one policy per uploaded file.

The PDF may contain selectable text or readable scanned pages.

Release 1 should define and display reasonable file and page limits. A limit around ten pages is appropriate for the intended document, but the final value should be confirmed during architecture planning using the sample set.

The product should not claim reliable support for every carrier and every declaration format. It should be tested against a documented benchmark containing several different layouts.

Recommended first field set

The first field set should be useful for agency intake without attempting to represent every detail on an auto policy.

Policy identification

- insurance carrier name
- policy number
- effective date
- expiration date

These fields identify the policy and determine the period represented by the document.

Named insured

- primary named insured full name
- insured mailing address

Additional named insureds may be captured as a list when clearly present, but complex person relationships are not required.

Premium

- total policy premium, when clearly stated

The product should not calculate a premium by adding unrelated charges. If a clear total cannot be supported, the field should remain missing or uncertain.

Vehicles

For each listed vehicle:

- year
- make
- model
- VIN

A partial VIN should not be represented as a complete VIN. The reviewer must be able to correct OCR mistakes easily.

Drivers

For each listed driver:

- full name

Release 1 should not extract driver license numbers, dates of birth, driver classifications, violation history, or detailed driver status unless later product research proves they are essential.

Core liability coverage

- bodily injury liability limit
- property damage liability limit

These are central personal auto coverage values and are useful during intake.

Optional coverage when clearly present

- uninsured or underinsured motorist limit
- comprehensive deductible by vehicle
- collision deductible by vehicle

These values should remain optional because policies and state requirements differ.

Fields to delay

The following should not be extracted in Release 1:

- agent or agency contact details
- discounts
- surcharges
- taxes and individual fees
- payment plan details
- lienholders
- endorsements
- policy form numbers
- rating information
- garaging details
- driver license numbers
- dates of birth
- accident and violation history
- rental reimbursement
- towing or roadside assistance
- medical payments
- personal injury protection details
- state-specific specialist fields
- coverage recommendations
- inferred gaps in coverage

These fields increase variability and review complexity without being necessary to prove the core product.

9. Human review experience

The review experience should be designed around one question:

Can the reviewer quickly understand what the AI found, where it found it, what may be wrong, and what will happen after approval?

Review header

The top of the page should show:

- customer
- document name
- upload time
- current status
- processing method
- number of warnings
- most recent processing attempt
- whether this may be a duplicate

Source document

The reviewer should be able to read the uploaded PDF without downloading it.

Selecting a field should take the reviewer to the relevant page when page evidence exists. A supporting text excerpt should be displayed where available.

Exact visual bounding boxes are attractive but should not be mandatory for Release 1 if they require expensive or provider-specific document analysis. Page navigation and evidence text are enough for a believable first release.

Field organization

Fields should be divided into understandable groups:

- policy
- named insured
- premium
- vehicles
- drivers
- coverage

Each group should show whether it contains unresolved warnings.

Field state

A field should communicate more than a value. It should also show a simple state such as:

- extracted with evidence
- extracted with weak evidence
- missing
- invalid
- conflicting
- affected by OCR quality
- corrected by reviewer
- marked not present

Corrections

When a reviewer changes a value:

- the original extracted value remains available in history
- the new value is clearly marked as a reviewer correction
- the reviewer does not need to write a note for every spelling correction
- important corrections may allow an optional reason
- the correction does not rewrite the original AI response

Approval

Before approval, the product should summarize:

- fields that will update the customer record
- unresolved warnings
- reviewer corrections
- information that remains missing

The reviewer should be required to resolve or acknowledge critical warnings.

Approval is a business action. It should be clear and deliberate, not an automatic result of opening or saving the page.

Rejection

The reviewer should select a simple reason such as:

- unsupported document
- unreadable document
- wrong customer
- wrong document type
- duplicate document
- information cannot be confirmed
- other

Rejected documents remain visible in history but do not update the active policy record.

Reprocessing

Reprocessing should be available for appropriate failures or after an extraction configuration change.

It should not be used as an unlimited “try again” button. Each attempt can cost money and may produce a different result.

The reviewer should understand:

- why reprocessing is available
- whether it will use a paid service
- which previous result will remain preserved
- that approved customer information will not be overwritten automatically

Review tracking

The product should record:

- who reviewed the document
- when review began
- what was changed
- which warnings were acknowledged
- whether the document was approved or rejected
- the final approved values

Release 1 does not need reviewer assignment, comments, mentions, manager approval, or simultaneous editing.

10. AI uncertainty and failure handling

The product must not claim that AI “knows” a value

The AI produces a candidate interpretation of the document. The product then combines that candidate with evidence and validation signals.

A model saying it is “95% confident” does not make the result 95% correct. Model self-confidence should be treated as a weak signal, not a decision.

Better uncertainty signals

The reviewer’s level of trust should be influenced by several observable signals:

- supporting text exists in the document
- the supporting text is located on a known page
- the value has the correct format
- dates are in a reasonable order
- a VIN has a valid-looking structure
- a monetary value is labelled as a total premium
- related values agree
- OCR quality is acceptable
- the model did not return several competing values
- required policy identity fields are present
- a second extraction method agrees, when such a method is used
- previous reviewer corrections reveal a repeated problem with a layout or field

Release 1 does not need a complex mathematical scoring system. It needs clear, explainable warning reasons.

Unreadable documents

When the PDF cannot be opened or pages cannot be read:

- processing stops safely
- no customer policy update occurs

- the user receives a clear reason
- the document can be rejected or replaced
- expensive extraction calls should be avoided when failure can be detected earlier

Scanned documents

When usable embedded text is absent:

- OCR is attempted
- the product records that OCR was used
- low OCR quality creates visible warnings
- extracted values remain unapproved

Rotated or low-quality pages

The system should attempt reasonable handling where supported. When quality remains poor, it should flag the document instead of hiding the issue.

Release 1 does not need advanced image restoration.

Missing fields

A missing field should remain missing.

The model must not fill gaps using assumptions about common insurance policies.

The reviewer may mark the value as not present in the document.

Conflicting values

When several possible policy numbers, dates, premiums, or limits are detected:

- the product should not silently choose one
- the conflict should be visible
- the reviewer should compare the candidates with the source

Unsupported document types

A home declaration, insurance card, claim form, invoice, or unrelated document should not be processed as though it were a personal auto declaration.

The system should stop or mark it unsupported before approval.

Invalid dates or monetary values

Invalid formats or impossible relationships should create warnings.

Examples include:

- expiration before effective date
- malformed dates

- a value labelled as a premium that cannot be interpreted safely
- conflicting totals

The system should not automatically invent corrected values.

Invalid AI structure

When the AI response does not match the required result structure:

- it should not enter the review queue as a normal successful result
- the system may make a limited controlled repair attempt
- repeated failure should produce a visible failed status
- the raw failure information should be available for technical diagnosis without exposing sensitive details to normal users

Provider errors and timeouts

Provider failures should:

- create a recoverable failed or delayed state
- avoid creating partial approved data
- support a limited retry
- preserve the original document
- avoid charging repeatedly through uncontrolled retries
- display a user-friendly message

Duplicate processing and repeated uploads

The system should identify exact or obvious duplicates before starting another paid workflow.

A reviewer may still intentionally reprocess a document, but the decision should be explicit and recorded.

Reviewer corrections

Reviewer corrections become the trusted result for that approved policy version.

Later AI attempts must not silently replace them.

Corrections should also be preserved as future evaluation data. Release 1 does not need automated learning from corrections, but it should retain enough context to study recurring errors later.

Model and prompt changes

Every processing attempt should record which extraction configuration produced it.

When a model or instruction changes:

- old results remain understandable
- benchmark documents can be processed again for comparison
- a new attempt does not erase the old attempt

- approved customer records are not automatically changed
-

11. Release 1 quality bar

Release 1 is ready for a public portfolio demonstration only when the following conditions are satisfied.

Workflow completion

- A non-developer can sign in, create or select a customer, upload a supported document, review the result, make corrections, approve it, and view the updated customer record.
- The workflow requires no database editing, command-line action, or developer intervention.
- A user can reject or safely retry an unsuccessful document.
- The user always understands the next available action.

Benchmark documents

A controlled benchmark should contain at least:

- several different declaration page layouts
- text-based PDFs
- readable scanned PDFs
- a rotated or difficult page
- a document with missing optional fields
- a document containing multiple vehicles
- a document containing multiple drivers
- an unsupported document
- an exact duplicate
- a deliberately poor-quality document

Synthetic documents should be used for the public demo. Private real-world samples may be used only when permission and secure handling are clear.

Extraction quality

For the approved benchmark set:

- every critical policy identity field must either be correct or visibly flagged for review
- there must be no silently accepted unsupported critical value
- clean text PDFs should meet an agreed high field-level correctness target
- readable scans may have a lower target, but mistakes must remain easy to detect and correct
- missing values must not be replaced with invented values
- multiple vehicles and drivers must not be merged together
- the same fixed extraction version should produce acceptably repeatable results

A practical initial target is:

- at least 95% field correctness on the curated clean text-PDF benchmark
- at least 90% field correctness on the curated readable scanned-PDF benchmark
- zero silently wrong critical fields after the product's validation and warning layer is considered

These are internal release gates, not public marketing claims.

Processing behavior

- Processing begins without freezing the page.
- The interface shows progress immediately.
- The curated demonstration documents complete within a documented demonstration limit.
- Provider timeouts result in a clear recoverable state.
- Refreshing or closing the page does not lose the job.
- Retry behavior is limited and visible.

A reasonable demo target is for curated text PDFs to complete within approximately one minute and OCR documents within approximately two minutes under normal provider conditions. These targets should be validated before being promised publicly.

Review behavior

- Every uncertain or invalid field is visually distinct.
- The reviewer can find supporting evidence without searching every page manually.
- The original extracted value remains preserved after correction.
- Critical warnings must be resolved or acknowledged before approval.
- Rejection requires a reason.
- Review and approval actions identify the acting user.

Data integrity

- An unapproved extraction never changes active customer policy data.
- Repeated approval does not create repeated active records.
- A retry does not erase an approved result.
- An exact duplicate does not trigger an unnoticed paid workflow.
- Approved values update the intended customer.
- One customer's document cannot be attached to another customer without a visible action.
- Current and previous approved policy versions can be distinguished.

Auditability

- Every major workflow action appears in understandable history.
- The processing method and extraction version are recorded.
- Reviewer corrections are visible.
- The source document connected to approved data is identifiable.
- Failures and retries are preserved.

UX completeness

Each major screen includes designed:

- loading states
- empty states
- success states
- warning states
- failure states
- disabled states

- retry states

No core screen should expose raw provider errors or leave the user with an unexplained spinner.

Demo readiness

- Demo data contains no real customer personal information.
 - The sample workflow can be restored without engineering access.
 - At least one complete workflow is available using cached or stored results.
 - Live AI usage can be disabled.
 - Public users cannot generate unlimited paid processing.
 - Email is restricted to approved test destinations or safely simulated.
 - The demo remains useful during a temporary AI provider outage.
 - The product contains a clear demonstration disclaimer.
 - A complete Loom can be recorded without editing the database or restarting services.
-

12. Budget boundaries

General budget rule

Release 1 should pay only for capabilities that materially improve the golden path.

A service being inexpensive does not automatically justify adding it. Every external service adds configuration, failure modes, secrets, monitoring needs, and possible future price changes.

Likely cost-producing services

AI model usage

Structured extraction is likely to be the main usage-based development cost.

Cost can increase because of:

- long document text
- OCR text containing unnecessary noise
- repeated extraction attempts
- overly large instructions
- using a more expensive model for every document
- public demo abuse
- reprocessing benchmark documents too frequently

Controls

- use one primary model
- use a smaller capable model where benchmark quality supports it
- limit pages and file sizes
- remove irrelevant text before the model call where safe
- cache benchmark results
- support replay mode
- restrict live processing to authenticated or approved users

- cap attempts per document
- record usage by attempt
- run the full benchmark only after meaningful extraction changes

OCR

OCR may charge by page or document depending on the provider.

Controls

- attempt direct PDF text extraction first
- invoke OCR only when text quality is inadequate
- limit page count
- cache OCR results
- do not repeat OCR when testing only the AI extraction step
- keep reusable OCR output for benchmark documents
- stop early for unsupported or unreadable files

Backend hosting

A low-cost or free environment may be suitable during development. A real customer or dependable public demo may require paid compute for predictable availability and background processing.

Risk

Slow startup, unavailable background work, or limited compute can make a live demo appear unreliable.

Boundary

Use the lowest-cost environment that reliably supports development. Pay for predictable availability only when the public demo or first customer requires it.

Frontend hosting

The frontend is unlikely to be the main cost at low traffic, but bandwidth and usage limits should still be monitored.

The separation between Vercel, Render, and Supabase is acceptable for this portfolio product, but it means three platforms must be configured and monitored.

The architecture document should confirm that the operational benefit is worth the extra deployment boundaries.

PostgreSQL and storage

Supabase is suitable for the preferred direction, but cost can increase through:

- stored PDFs
- processed versions
- database growth

- backups
- file downloads
- public demo traffic
- long retention
- logs or audit history

Controls

- private document storage
- synthetic demo files
- file-size limits
- clear retention rules
- no unnecessary duplicate copies
- safe cleanup of abandoned demo uploads
- limited public file access
- monitoring of storage and outbound transfer

Email

A small number of transactional acknowledgements should be inexpensive, but public misuse can create cost and reputation problems.

Controls

- send only after approval
- restrict demo recipients
- rate-limit sending
- use one fixed template
- prevent arbitrary bulk sending
- record send outcomes
- allow demonstration without sending a real email

Monitoring and error tracking

Basic provider logs and application monitoring may begin with free or low-cost options.

Advanced log retention, session replay, alerting, and observability platforms are optional until traffic or a customer justifies them.

Domain and related setup

A custom domain is a predictable portfolio expense. It improves trust but does not require additional enterprise networking products.

Development cost boundary

During development:

- use synthetic declaration pages
- use a small benchmark set
- cache direct text extraction, OCR, and AI results separately

- allow later stages to be tested without repeating earlier paid stages
- run live providers intentionally rather than on every interface test
- use replayed processing results for frontend development
- set a hard development usage limit
- do not pay for voice or telephony
- do not maintain several model providers
- do not run large automated AI test suites on every source-control change

Live demo cost boundary

For the public demo:

- most sample documents should use stored results
- live AI should be clearly labelled and tightly limited
- a visitor should not be able to upload unlimited files
- document size and page limits should be strict
- the demo should have a daily or monthly usage ceiling
- email should be restricted
- synthetic documents should be reusable
- the system should be resettable
- always-on backend hosting should be paid for only when demo reliability requires it

Real customer cost boundary

A real agency would likely require:

- dependable backend availability
- dependable background processing
- production database and storage
- backups
- monitoring
- regular AI and OCR usage
- controlled email sending
- secure retention and deletion practices
- support for higher document volume

These should be treated as customer operating costs rather than portfolio-development costs.

Optional future infrastructure cost

The following should not be paid for during Release 1:

- telephony
- voice provider production capacity
- several background worker services
- multi-region infrastructure
- enterprise search
- data warehouse
- advanced analytics platform
- dedicated vector database
- event streaming platform
- multi-provider routing

- enterprise identity management
- subscription billing
- compliance certification programs
- high-volume email
- advanced document intelligence unless simpler OCR is inadequate

Costs most likely to increase unexpectedly

The biggest cost risks are:

- public upload abuse
- OCR applied to every page unnecessarily
- repeated AI retries
- large or image-heavy PDFs
- sending full raw documents to expensive models
- keeping every temporary file forever
- downloading large PDFs repeatedly
- verbose production logs containing document content
- always-on compute added too early
- voice and telephony added before usage is validated

13. Main product risks

Risk 1: The product becomes too broad

Why it matters

Document AI, Voice AI, CRM, audit, analytics, and communications can quickly become several products. A broad but incomplete release will look less credible than one focused workflow.

How it can be reduced

- choose the document-first strategy
- use one document type
- use one primary user
- maintain explicit exclusions
- require a scope decision before adding any feature
- judge every feature by whether it improves the golden path

Must it be solved in Release 1?

Yes. Scope control is required from the beginning.

Risk 2: The product solves a technical problem but not a real agency workflow

Why it matters

A polished extraction demo may still fail to match how agency staff actually receive, review, and use declaration pages.

How it can be reduced

- interview at least a small number of insurance professionals when possible
- show them the proposed field set and review flow
- learn what information they enter before quoting
- avoid claiming integration with systems that have not been validated
- keep the first workflow easy to modify after feedback

Must it be solved in Release 1?

Partly. Release 1 can be built as a portfolio product, but its workflow assumptions should be validated before presenting it as a proven commercial solution.

Risk 3: AI returns plausible but incorrect values

Why it matters

A believable wrong value may be harder to detect than a missing value. Policy numbers, dates, VINs, limits, and deductibles can affect later business work.

How it can be reduced

- require evidence
- validate formats and relationships
- flag conflicts
- prevent automatic customer updates
- require human approval
- retain reviewer corrections
- create a representative benchmark
- separate critical and optional fields

Must it be solved in Release 1?

Yes. This is the central AI safety requirement.

Risk 4: Declaration page layouts vary widely

Why it matters

A workflow that works on one sample may fail when fields appear on different pages, under different labels, or in different table structures.

How it can be reduced

- define a narrow document boundary

- test several layouts
- include multiple vehicles and drivers
- avoid excessive field coverage
- retain layout-specific failures for evaluation
- describe support honestly

Must it be solved in Release 1?

Yes, within the defined benchmark. Universal carrier support is not required.

Risk 5: OCR creates incorrect text

Why it matters

OCR may confuse letters and numbers, particularly in policy numbers, VINs, dates, and monetary values.

How it can be reduced

- use direct PDF text when available
- show when OCR was used
- validate high-risk formats
- provide source page evidence
- flag poor-quality scans
- make correction fast
- avoid claiming certainty from OCR alone

Must it be solved in Release 1?

Yes, for readable scanned PDFs. Advanced restoration of severely damaged documents is not required.

Risk 6: Human review is slower than manual entry

Why it matters

If the reviewer has to inspect every field and repeatedly search the PDF, the product may move work rather than reduce it.

How it can be reduced

- organize fields into logical groups
- prioritize warnings
- show evidence and page references
- make keyboard and editing behavior efficient
- keep the field set small
- test the workflow with timed internal exercises
- reduce attention on high-evidence fields without hiding them

Must it be solved in Release 1?

Yes. Review speed is a central product outcome.

Risk 7: Approved data is overwritten or duplicated

Why it matters

Retries, repeated uploads, and accidental double actions can corrupt the customer record or make it unclear which policy is current.

How it can be reduced

- separate extraction attempts from approved policy versions
- make approval deliberate
- detect duplicates
- design repeated actions to be safe
- preserve history
- require an explicit replacement of active approved data

Must it be solved in Release 1?

Yes.

Risk 8: External providers fail during a demonstration

Why it matters

AI and OCR providers may experience delays, timeouts, quota problems, or temporary outages. A failed Loom or client demo reduces trust.

How it can be reduced

- provide stored benchmark results
- support replay mode
- display clear failure states
- use limited controlled retries
- avoid making the complete demo dependent on live AI
- preserve completed sample workflows

Must it be solved in Release 1?

Yes for demonstrations. Multi-provider failover is not required.

Risk 9: Public demo users create uncontrolled cost

Why it matters

A visitor can repeatedly upload large documents or trigger paid processing.

How it can be reduced

- require controlled access
- use quotas
- restrict file sizes and page counts
- limit live processing
- use synthetic samples
- provide replayed results
- disable processing when a budget ceiling is reached

Must it be solved in Release 1?

Yes before public access.

Risk 10: Sensitive customer information is exposed

Why it matters

Declaration pages may contain names, addresses, VINs, policy numbers, and other personal information.

How it can be reduced

- use private file storage
- require authentication
- prevent public file URLs
- use synthetic data in demos
- control document retention
- avoid sensitive information in normal logs
- define deletion behavior
- send documents only to approved processing providers
- clearly separate development, demo, and customer environments

Must it be solved in Release 1?

Baseline protection must be solved. Formal enterprise certifications and complex compliance programs are outside Release 1.

Risk 11: The product appears to verify insurance coverage

Why it matters

Reading a declaration page does not confirm that a policy is active, paid, unchanged, or valid with the carrier.

How it can be reduced

- use careful interface language

- describe results as extracted and reviewer-approved
- avoid “verified policy” unless carrier verification exists
- keep confirmation emails narrow
- include an appropriate product disclaimer

Must it be solved in Release 1?

Yes.

Risk 12: Three hosting platforms increase operational complexity

Why it matters

Using Vercel, Render, and Supabase creates separate environments, credentials, configuration, monitoring, and failure points.

How it can be reduced

- keep responsibilities clear
- avoid adding more platforms without need
- maintain straightforward environment configuration
- centralize operational documentation
- verify that cross-platform communication is secure and reliable
- reconsider consolidation only if the architecture document finds a meaningful benefit

Must it be solved in Release 1?

It must be understood and documented. Platform consolidation is not automatically required.

Risk 13: Provider abstraction becomes over-engineered

Why it matters

Trying to make every AI and OCR provider interchangeable can create a large internal framework before the product has one reliable workflow.

How it can be reduced

- use one production provider
- define a clean boundary around extraction
- store normalized internal results
- preserve raw provider responses securely for diagnosis when appropriate
- avoid provider-selection screens and multi-provider routing

Must it be solved in Release 1?

A clean boundary is required. Full provider interchangeability is not.

Risk 14: The benchmark becomes easier than real documents

Why it matters

The product may appear accurate because development repeatedly uses the same clean samples.

How it can be reduced

- keep a separate evaluation set
- include difficult layouts and scans
- avoid changing prompts only to fix one document
- record failures by document and field
- add new samples over time
- do not publicly claim general accuracy from a small internal benchmark

Must it be solved in Release 1?

Yes at a basic level.

14. Open decisions

The following decisions require approval before architecture and implementation planning should continue.

Decision 1: Primary user and use case

Recommended approval

Build for a new-business intake specialist or personal-lines account manager at a small independent U.S. insurance agency who processes a prospect's current auto declaration page before quoting or onboarding.

Approval confirms that Release 1 is not for carriers, brokers of every type, claims departments, enterprise agencies, or general document processing.

Decision 2: Release strategy

Recommended approval

Choose the document-first strategy and move browser Voice AI to Release 2.

Approval confirms that no voice interface, transcription, speech output, or telephony work will enter Release 1.

Decision 3: Supported document boundary

Recommended approval

Support English-language U.S. personal auto declaration page PDFs containing one policy per file.

Approval confirms that Release 1 will not claim support for other countries, languages, policy types, or arbitrary insurance documents.

Decision 4: First field set

Recommended approval

Use the field groups defined in Section 8:

- policy identity
- named insured and address
- total premium when present
- vehicles
- driver names
- bodily injury and property damage liability
- optional uninsured or underinsured motorist limits
- optional comprehensive and collision deductibles

Approval confirms that the delayed fields are intentionally excluded.

Decision 5: Customer confirmation

Recommended approval

Include one reviewer-triggered acknowledgement email after approval, with strict demo restrictions.

Approval confirms that this is a narrow transactional action and not the beginning of a messaging platform.

Decision 6: Public demo access model

Recommended approval

Use a controlled demonstration account or request-access flow with synthetic data, replayable sample results, and tightly limited live AI processing.

Approval confirms that anonymous visitors will not receive unlimited document uploads or paid AI access.

15. Recommended next planning document

After this Release 1 definition is approved, the next document should be:

“Release 1 System Architecture and Technical Decision Record”

That document should define:

- modular monolith boundaries
- ownership of customer, document, review, audit, and usage responsibilities
- document lifecycle and processing states
- background processing approach
- extraction and OCR provider decision
- model provider decision
- internal provider boundaries
- approved-data versioning
- retry and duplicate-safety principles
- authentication and document security
- deployment responsibilities across Vercel, Render, and Supabase
- development, demo, and customer environments
- cost-control mechanisms
- observability and failure diagnosis
- alternatives considered
- accepted trade-offs
- the first decision log
- human approval checkpoints before Codex implements each major architectural area

Implementation tasks, repository structure, APIs, database design, and coding prompts should be created only after that architecture document is reviewed and approved.